ensemble

April 27, 2025

[1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv("dt_prepared.csv")

X = df.drop("Target", axis=1)
y = df["Target"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=11, stratify=y
)

print("Training Set Shape:", X_train.shape)
print("Testing Set Shape:", X_test.shape)

X_train.head(100).to_csv("ensemble_train_sample.csv", index=False)
X_test.head(100).to_csv("ensemble_test_sample.csv", index=False)

# Display previews
print("\nTrain Data Sample:")
display(X_train.head())

print("\nTest Data Sample:")
display(X_test.head())
```

```
Training Set Shape: (56000, 33)
Testing Set Shape: (14000, 33)

Train Data Sample:

       Genetic Markers  Autoantibodies  Family History  Environmental Factors  \
46981                0               0               0                      1
53951                1               0               0                      0
17370                1               1               1                      1
35377                0               0               1                      1
4467                 1               1               1                      0

       Insulin Levels  Age   BMI  Physical Activity  Dietary Habits  \
46981            23.0   37  27.0                  2               1
```

```
53951            33.0   14  23.0                  0                1
17370             8.0   24  16.0                  2                0
35377            11.0   13  15.0                  2                1
4467              9.0   10  14.0                  1                1

       Blood Pressure  …  Pulmonary Function  Cystic Fibrosis Diagnosis  \
46981             121  …                  78                          0
53951             103  …                  50                          1
17370              91  …                  84                          0
35377             106  …                  78                          1
4467               72  …                  55                          1

       Steroid Use History  Genetic Testing  Neurological Assessments  \
46981                    0                1                         1
53951                    0                1                         2
17370                    0                1                         2
35377                    1                0                         2
4467                     1                0                         1

       Liver Function Tests  Digestive Enzyme Levels  Urine Test  \
46981                     0                       44           2
53951                     0                       73           2
17370                     1                       41           3
35377                     1                       55           2
4467                      1                       27           2

       Birth Weight  Early Onset Symptoms
46981          3653                     0
53951          3159                     0
17370          2558                     1
35377          3972                     1
4467           1914                     1

[5 rows x 33 columns]


Test Data Sample:

       Genetic Markers  Autoantibodies  Family History  Environmental Factors  \
4350                 1               1               0                      0
14665                0               0               1                      0
57348                0               1               1                      0
65011                1               1               0                      0
62793                0               0               1                      1

       Insulin Levels  Age   BMI  Physical Activity  Dietary Habits  \
4350             19.0   39  25.0                  2               0
14665            15.0   53  30.0                  2               0
57348            16.0   35  31.0                  0               1
```

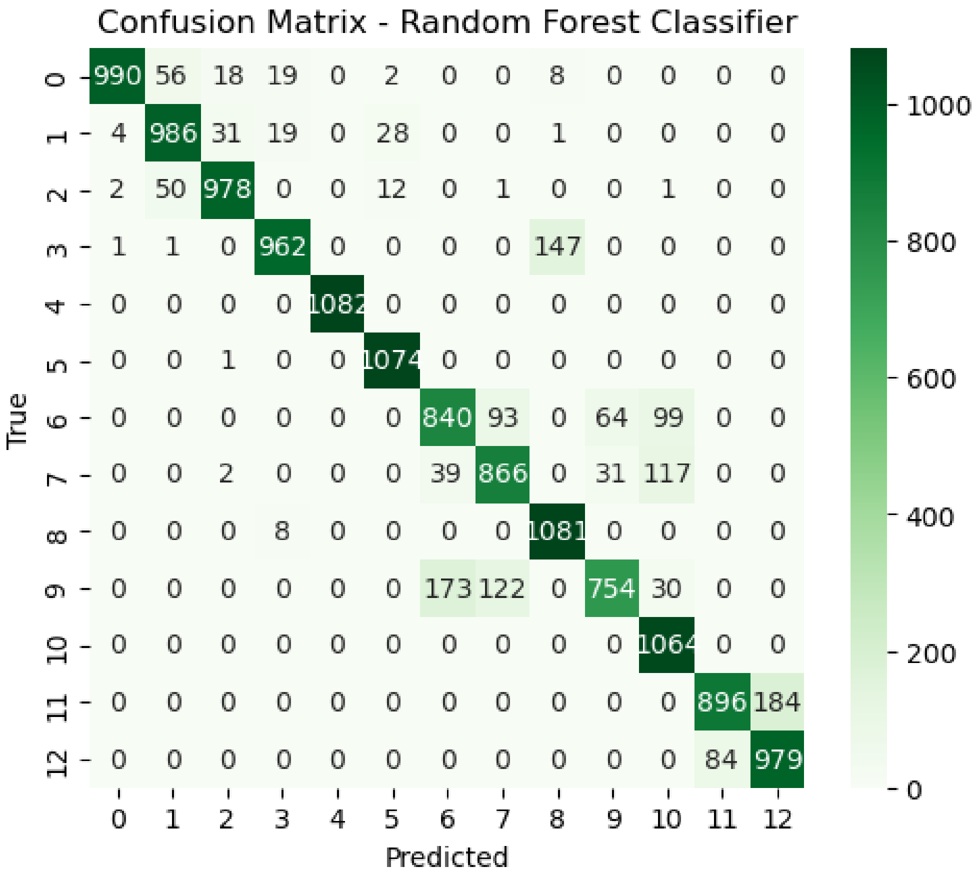
```
65011             12.0   21  17.0                0               1
62793             27.0   44  26.0                0               0

       Blood Pressure  …  Pulmonary Function  Cystic Fibrosis Diagnosis  \
4350              120  …                  78                          0
14665             126  …                  72                          0
57348             119  …                  81                          1
65011             106  …                  77                          1
62793             140  …                  65                          0

       Steroid Use History  Genetic Testing  Neurological Assessments  \
4350                     0                1                         2
14665                    1                1                         2
57348                    1                0                         2
65011                    0                1                         2
62793                    0                0                         2

       Liver Function Tests  Digestive Enzyme Levels  Urine Test  \
4350                      0                       68           2
14665                     1                       77           2
57348                     0                       72           0
65011                     1                       63           3
62793                     1                       73           1

       Birth Weight  Early Onset Symptoms
4350           3294                     1
14665          3898                     1
57348          4057                     1
65011          2810                     0
62793          3548                     1

[5 rows x 33 columns]
```

[2]:

```python
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

y_pred = rf_model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(f"Random Forest Accuracy: {accuracy:.4f}\n")
```

```python
print("Classification Report:")
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Greens')
plt.title("Confusion Matrix - Random Forest Classifier")
plt.xlabel("Predicted")
plt.ylabel("True")
plt.show()
```

```
Random Forest Accuracy: 0.8966

Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.91      0.95      1093
           1       0.90      0.92      0.91      1069
           2       0.95      0.94      0.94      1044
           3       0.95      0.87      0.91      1111
           4       1.00      1.00      1.00      1082
           5       0.96      1.00      0.98      1075
           6       0.80      0.77      0.78      1096
           7       0.80      0.82      0.81      1055
           8       0.87      0.99      0.93      1089
           9       0.89      0.70      0.78      1079
          10       0.81      1.00      0.90      1064
          11       0.91      0.83      0.87      1080
          12       0.84      0.92      0.88      1063

    accuracy                           0.90     14000
   macro avg       0.90      0.90      0.90     14000
weighted avg       0.90      0.90      0.90     14000
```


Confusion Matrix - Random Forest Classifier
990 56 18 19 0 2 0 0 8 0 0 0 0
4 986 31 19 0 28 0 0 1 0 0 0 0
2 50 978 0 0 12 0 1 0 0 1 0 0
1 1 0 962 0 0 0 0 147 0 0 0 0
0 0 0 0 1082 0 0 0 0 0 0 0 0
0 0 1 0 0 1074 0 0 0 0 0 0 0
0 0 0 0 0 0 840 93 0 64 99 0 0
0 0 2 0 0 0 39 866 0 31 117 0 0
0 0 0 8 0 0 0 0 1081 0 0 0 0
0 0 0 0 0 0 173 122 0 754 30 0 0
0 0 0 0 0 0 0 0 0 0 1064 0 0
0 0 0 0 0 0 0 0 0 0 0 896 184
0 0 0 0 0 0 0 0 0 0 0 84 979
True
0 1 2 3 4 5 6 7 8 9 10 11 12
Predicted
1000
800
600
400
200
0

[ ]: